

RFEM ↔ Karamba3D

Results Validation - Wind Loading

1 Document Information

Project Name	Steel Tower Example
Model / Example	01_Integrated Steel Tower
Software Versions	RFEM6 (Dlubal) 6.13.0003 Karamba3D 3.1.60410.0
Author	Dr Dragos Naicu
Date	07.05.2026
Document Version	v01

2 Model Overview

Structural System	Steel Frame
Geometry	Rectangular plan and regular in height
Size	6.0m x 6.0m plan with 3 levels of 2.4m height
Material	Steel
Cross-Sections	HEA, IPE, SHS and Round
Lateral System X	Moment resisting frame
Lateral System Y	Braced frame
Number of Members	193
Number of Nodes	117
Software	RFEM6 (Dlubal) ↔ Karamba3D 3.1.60410.0 Rhino 8 SR29 8.29.26063.11001
Model Source	Model originally based on RFEM from here: https://www.dlubal.com/en/downloads-and-information/examples-and-tutorials/models-to-download/005534 Model recreated in Karamba3D and then exported to RFEM via SAF Export. Validation / Comparison is not performed on the original RFEM, but on the exported version.
Validation Purposes	1. Validate wind pressures generated by C# script according to EC1 2. Validate results based on those wind loads

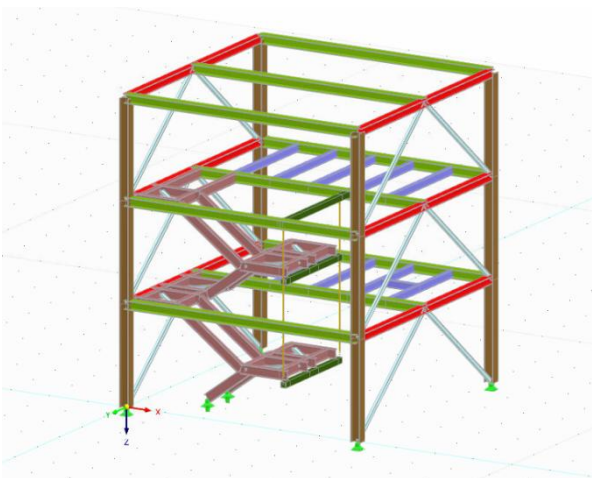
The example file produced uses a C# script to calculate **wind pressures, coefficients and design loads** parametrically, given the size and shape of the rectangular tower. This follows the approach from Section 7.2.2 of the Eurocode for *“Vertical walls of rectangular plan buildings”*. The derived pressure zones and values are then converted and applied as mesh loads in Karamba3D onto the structure. The main aim is to produce a detailed analysis model, with code compliant wind loads and to compare both the loading and the analysis results with a third-party software (e.g. Dlubal RFEM).

2.1 Modelling Deviations from Original RFEM Model

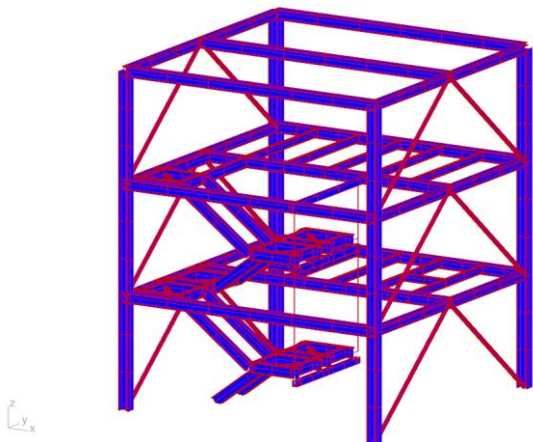
The following simplifications were made between the original model and the recreated model:

Original RFEM Feature	Description	Modification for Karamba3D
Steel Joints Add-On	The initial model had Steel Joints add-on activated and joints had been defined with spring rotational stiffnesses automatically in RFEM.	- joints were removed and simple pinned joints were implemented in both RFEM and Karamba3D. - this had the effect of making the model a little more sensitive/unstable for certain lateral directions. - the non-braced direction was modified to have moment resisting joints for the primary beams at the columns → in one direction, there are diagonal braces and in the other direction there are effectively moment frames.
Wind Load Wizard	The initial model uses Wind Load Wizard to automatically derive area loads and member load distributions according to EC1 and was using the German National Annex procedure.	a separate wind load wizard was created for validation purposes that used the basic EC1 procedure, without NA contributions. This was applied in the exported model.
Eccentricities	The initial model had quite a few eccentricities applied (primarily I think to create steelwork that is flush to top flange levels of primary beams).	this was not replicated in K3D

2.2 Model Screenshots



RFEM6 model - final validation configuration



Karamba3D model - final validation configuration

3 Wind Loading

Wind loads are derived according to EN 1991-1-4 using the basic EC1 procedure. In Karamba3d a basic wind pressure is used as input, whereas in RFEM a combination of basic wind speed and air density is used to derive the same value of basic wind pressure.

3.1 Design Parameters

Norm / Standard	BS EN 1991-1-4:2005+A1:2010	
Basic Wind Pressure q_b	0.320 kN/m ²	Input for Karamba3D
Air Density	1.25 kg/m ³	Input for RFEM
Fundamental Wind Velocity $v_{b,0}$	22.5 m/s	Input for RFEM
Terrain Category	Category 0	
Orography factor c_0	1.0	
Turbulence factor I	1.0	
Reference Height z_e	Generally $z_e = 7.20$ m (height of model)	
Peak Velocity Pressure q_p	Generally $q_p = 0.896$ kN/m ²	
Wind Directions modelled	+X and +Y (Rhino axes)	
Geometric Bounding Box	6.0m x 6.0m x 7.2m high	
Mesh resolution (K3D)	0.1m	

3.2 Calculation Assumptions

1. Geometry
 - a. Rectangular enclosed building (Section 7.2)
 - b. Bounding box approximation: real geometry may differ, zones computed for enclosing rectangle
 - c. Building origin at box corner[0]
2. Wind Climate
 - a. Flat terrain: $c_0(z) = 1.0$ (no orography)
 - b. Turbulence factor $k_l = 1.0$ (recommended)
 - c. Air density $\rho = 1.25$ kg/m³ (sea level, in q_b)
 - d. No altitude correction applied
 - e. Cardinal wind directions only (no diagonal)
 - f. Perpendicular-to-face governs (Section 7.2)
3. Pressure Profile
 - a. Standard EN logarithmic profile (Cl. 4.3-4.5)
 - b. Not a National Annex variant (e.g. DIN power-law)
 - c. $q_p(z) = c_e(z) * q_b$
 - d. $c_e(z) = k_r^2 * [\ln^2(z/z_0) + 7 * \ln(z/z_0)]$
 - e. z clamped to $\max(z, z_{min})$ per Cl. 4.3.2
 - f. Zones A, B, C, E: $z_e = h$ (Cl. 7.2.2(1) Note)
 - g. Zone D strips: $z_e = \text{top of each strip}$
4. Coefficients
 - a. $c_{pe,10}$ throughout (loaded area ≥ 10 m²)
 - b. No $c_{pe,1}$ correction for small areas

- c. Linear interpolation of Table 7.1 for D and E
- d. h/d clamped to range [0.25, 5.0]
- 5. Zone Geometry
 - a. Side face widths per Figure 7.5
 - b. w_B capped to $d - w_A$ when $d < e$ (Figure 7.5)
 - c. Middle zone ($h > 2b$) divided into equal strips
 - d. (user-specified, not prescribed by code)
- 6. Not Included
 - a. Structural factor c_{scd} (taken as 1.0)
 - b. Internal pressure c_{pi} (external only)
 - c. Torsional loading (Cl. 7.1.2)

3.3 Wind Pressure Validation

Due to the decimal precision display in RFEM, the exact deviation is not possible to ascertain but it is surely within rounding of the third decimal point. Therefore, in both absolute and relative terms it is very small.

Wind Pressure			
Result / Component	RFEM	Karamba3D	Deviation
Basic Wind Pressure q_b	0.32 kN/m ²	0.32 kN/m ²	0
Terrain Factor k_r	0.16	0.156	Rounding
Peak Velocity Pressure q_p in +X	0.89 kN/m ²	0.896 kN/m ²	Rounding
Peak Velocity Pressure q_p in +Y	0.89 kN/m ²	0.896 kN/m ²	Rounding

No. 1 Name Buildings | Vertical walls with flat/monopitch roof

Main Parameters Load Cases Results Tolerances

Parameters EN 1991 | CEN | 2015-09

Description	Symbol	Value	Unit
Terrain category		Category 0	
Structure height	h	7.200 m	
Air density	ρ	1.25 kg/m ³	
Fundamental wind velocity	$v_{b,0}$	22.50 m/s	
Orography factor	c_o	1.00 --	
Directional factor	c_{dir}	1.00 --	
Seasonal factor	c_{season}	1.00 --	
Turbulence factor	k_t	1.00 --	
Manual definition of terrain factor		☐	
Terrain factor	k_r	0.16 --	
Basic velocity pressure	q_b	0.32 kN/m ²	
Load distribution type		Trapezoidal	
Lack of correlation acc. to 7.2.2(3)		☐	
Lock for new objects		☐	
Consider member eccentricity		☐	
Consider cross-section distribution		☐	
Consider internal pressure		☐	

Wind Loading Information

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WALL ZONES – EN 1991-1-4 Section 7.2

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--- Box Geometry ---

Origin : (0.00, 0.00, 0.00)

b1 (along X) : 6.00 m

b2 (along Y) : 6.00 m

h : 7.20 m

--- Wind Direction ---

Direction : +Y

b (crosswind) : 6.00 m

d (along-wind) : 6.00 m

--- Terrain (Table 4.1) ---

Category : 0

z0 : 0.003 m

zmin : 1 m

k_r : 0.1560

--- Pressure ---

q_b : 0.320 kN/m²

$q_p(h)$: 0.896 kN/m²

--- Zone Geometry (Figure 7.5) ---

h/d : 1.200

$e = \min(b, 2h)$: 6.00 m

Zone A width : $e/5 = 1.20$ m

Zone B width : $d - w_A = 4.80$ m

Zone C : does not exist ($d \leq e$)

--- Coefficients $c_{pe,10}$ (Table 7.1) ---

Zone A : -1.20

Zone B : -0.80

Zone C : N/A

Zone D : 0.80

Zone E : -0.51

3.4 EC1 Pressure Zone Validation

EN 1991-1-4 prescribes 5 pressure zones A, B, C, D, E on vertical walls of rectangular buildings.

When considering the direction of the wind flow, zones A, B and C are symmetrical and refer to the sides of the building, whereas zone D is the windward face (front face) and zone E is the leeward face (behind face).

For faces A, B, C and E the reference height is the top of the building and there is one peak pressure.

For face D the reference height is based on the number of strips into which the zone is divided. This depends on the overall geometry of the building and follows a logarithmic profile in the code. For this particular case the zone D profile has 2 strips.

The final wind pressure is calculated as: $w_e = q_p \times C_{pe}$ and negative values indicate suction away from the building face, whereas positive values indicate pressure on the building face.

Pressure Zones: Direction +X and +Y (plan symmetry makes geometry identical)

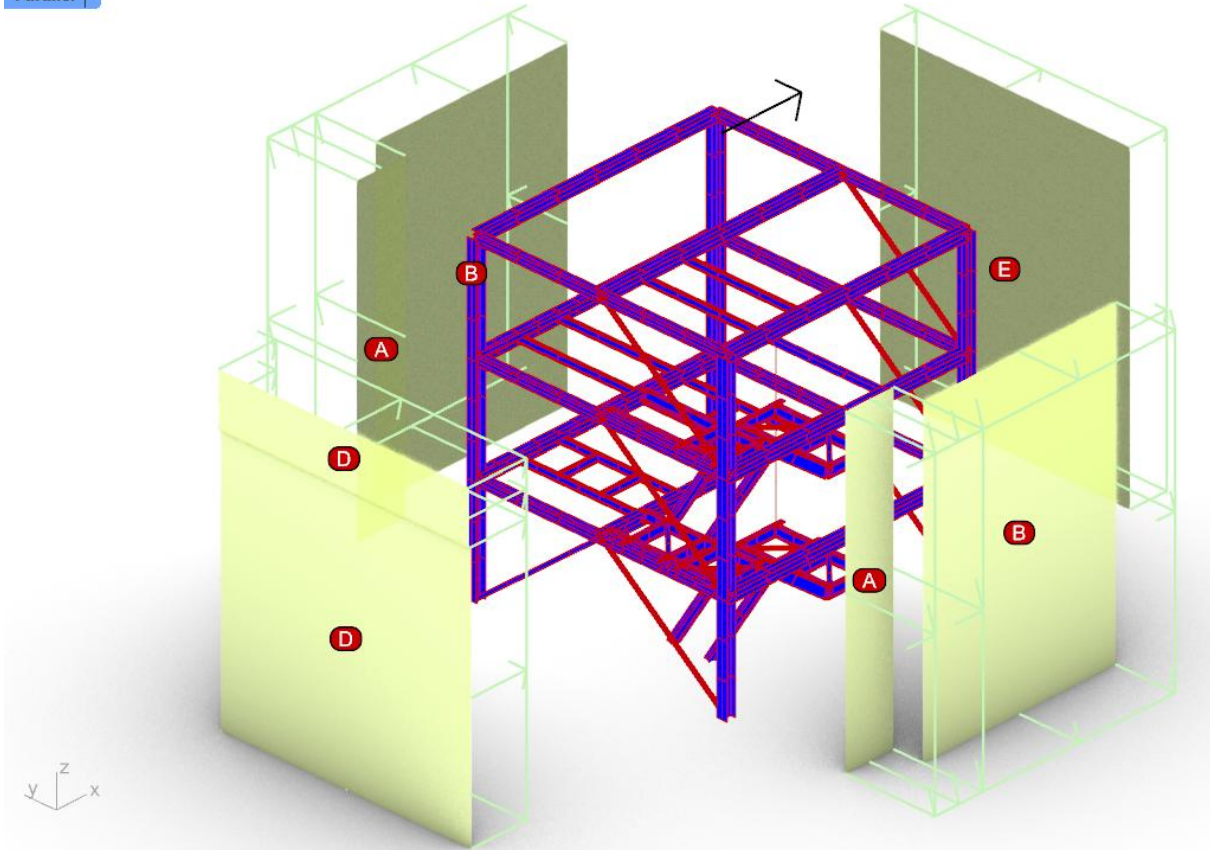
Zone	Exists?	Strips [#]	Ref Height z_e [kN/m ²]	Peak Pressure q_p [kN/m ²]	RFEM Pressure Coefficient C_{pe} [-]	Karamba3D Pressure Coefficient C_{pe} [-]	RFEM Design Wind Pressure w_e [kN/m ²]	Karamba3D Design Wind Pressure w_e [kN/m ²]	Deviation Absolute / %
A	Yes	1	7.2m	0.896	- 1.20	- 1.20	-1.06	-1.076	0.016 / 1.5%
B	Yes	1	7.2m	0.896	- 0.80	- 0.80	-0.71	-0.717	0.007 / 1.0%
C	No	-	-	-	-	-	-	-	-
D	Yes	2	6.0m 7.2m	0.865 0.896	0.80 0.80	0.80 0.80	0.68 0.71	0.692 0.717	0.012 / 1.8% 0.007 / 1.0%
E	Yes	1	7.2m	0.896	- 0.51	- 0.51	-0.45	-0.457	0.007 / 1.5%

Left: Design Pressure Results in +X; Right: Design Pressure Results in +Y

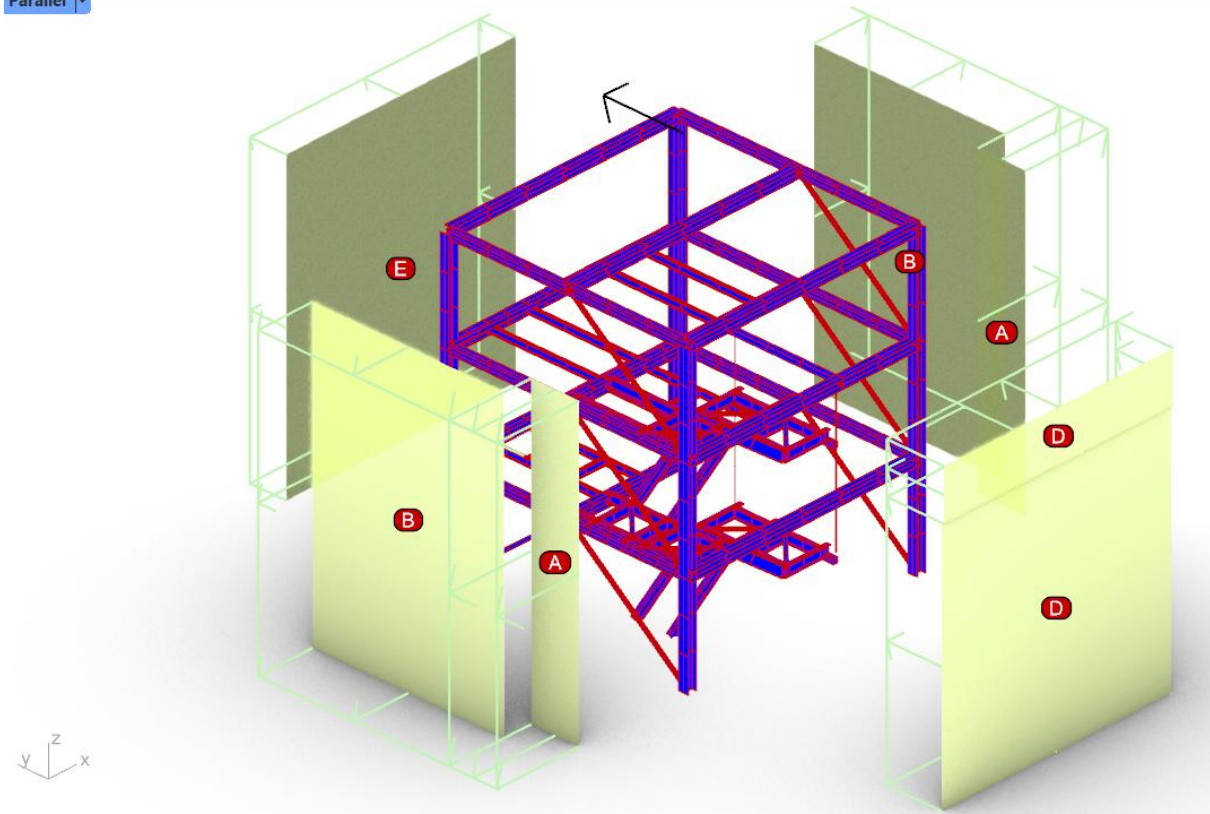
LC6 - Wind +X (+X Rh)				LCS - Wind -Y (-Y Rh)			
Building dimensions				Building dimensions			
Structure height	h	7.200 m		Structure height	h	7.200 m	
Structure width	b1	6.000 m		Structure width	b1	6.000 m	
Structure width	b2	6.000 m		Structure width	b2	6.000 m	
Structure depth	d1	6.000 m		Structure depth	d1	6.000 m	
Structure depth	d2	6.000 m		Structure depth	d2	6.000 m	
Split distance	e1	6.000 m		Split distance	e1	6.000 m	
Split distance	e2	6.000 m		Split distance	e2	6.000 m	
Split distance	e	6.000 m		Split distance	e	6.000 m	
Zone dimensions				Zone dimensions			
Zone A depth	da,1	1.200 m		Zone A depth	da,1	1.200 m	
Zone A depth	da,2	1.200 m		Zone A depth	da,2	1.200 m	
Zone B depth	ds,1	4.800 m		Zone B depth	ds,1	4.800 m	
Zone B depth	ds,2	4.800 m		Zone B depth	ds,2	4.800 m	
Velocity pressure				Velocity pressure			
Peak velocity pressure	q _p (h)	0.89 kN/m ²		Peak velocity pressure	q _p (h)	0.89 kN/m ²	
Zone A				Zone A			
External pressure coefficient	C _{pe,1}	-1.20 --		External pressure coefficient	C _{pe,1}	-1.20 --	
External pressure coefficient	C _{pe,2}	-1.20 --		External pressure coefficient	C _{pe,2}	-1.20 --	
External wind pressure	W _{e,1}	-1.06 kN/m ²		External wind pressure	W _{e,1}	-1.06 kN/m ²	
External wind pressure	W _{e,2}	-1.06 kN/m ²		External wind pressure	W _{e,2}	-1.06 kN/m ²	
Zone B				Zone B			
External pressure coefficient	C _{pe,1}	-0.80 --		External pressure coefficient	C _{pe,1}	-0.80 --	
External pressure coefficient	C _{pe,2}	-0.80 --		External pressure coefficient	C _{pe,2}	-0.80 --	
External wind pressure	W _{e,1}	-0.71 kN/m ²		External wind pressure	W _{e,1}	-0.71 kN/m ²	
External wind pressure	W _{e,2}	-0.71 kN/m ²		External wind pressure	W _{e,2}	-0.71 kN/m ²	
Zone D				Zone D			
External pressure coefficient	C _{pe}	0.80 --		External pressure coefficient	C _{pe}	0.80 --	
External wind pressure	W _e	0.71 kN/m ²		External wind pressure	W _e	0.71 kN/m ²	
External wind pressure	W _e (b)	0.68 kN/m ²		External wind pressure	W _e (b)	0.68 kN/m ²	
Zone E				Zone E			
External pressure coefficient	C _{pe}	-0.51 --		External pressure coefficient	C _{pe}	-0.51 --	
External wind pressure	W _e	-0.45 kN/m ²		External wind pressure	W _e	-0.45 kN/m ²	

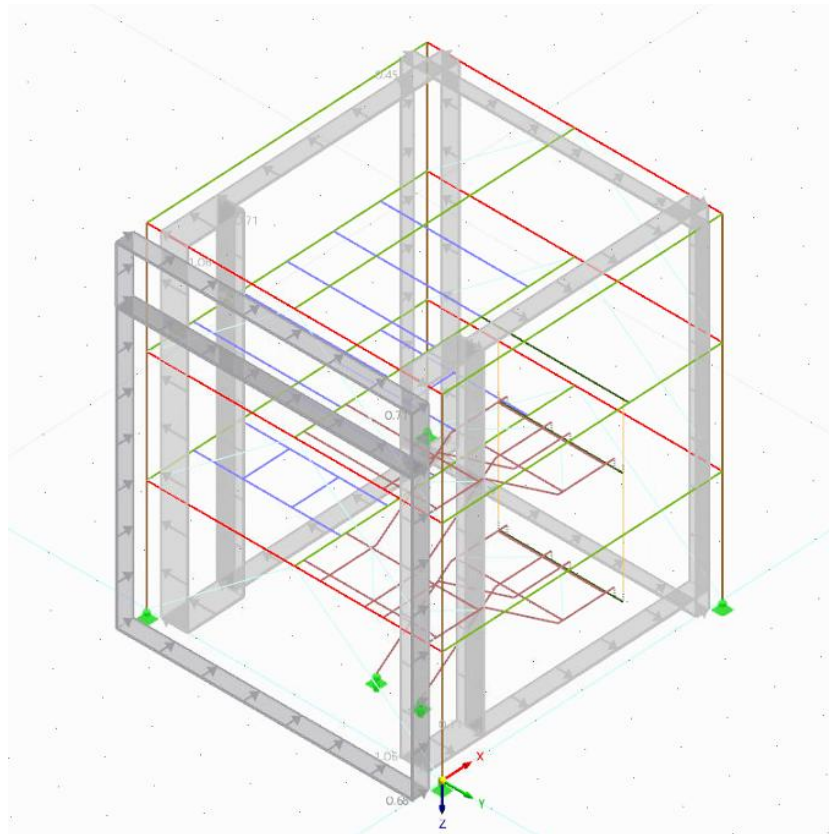
The following pages show visualisations of the determined zones in Rhino + Karamba3D and in RFEM.

Parallel ▾

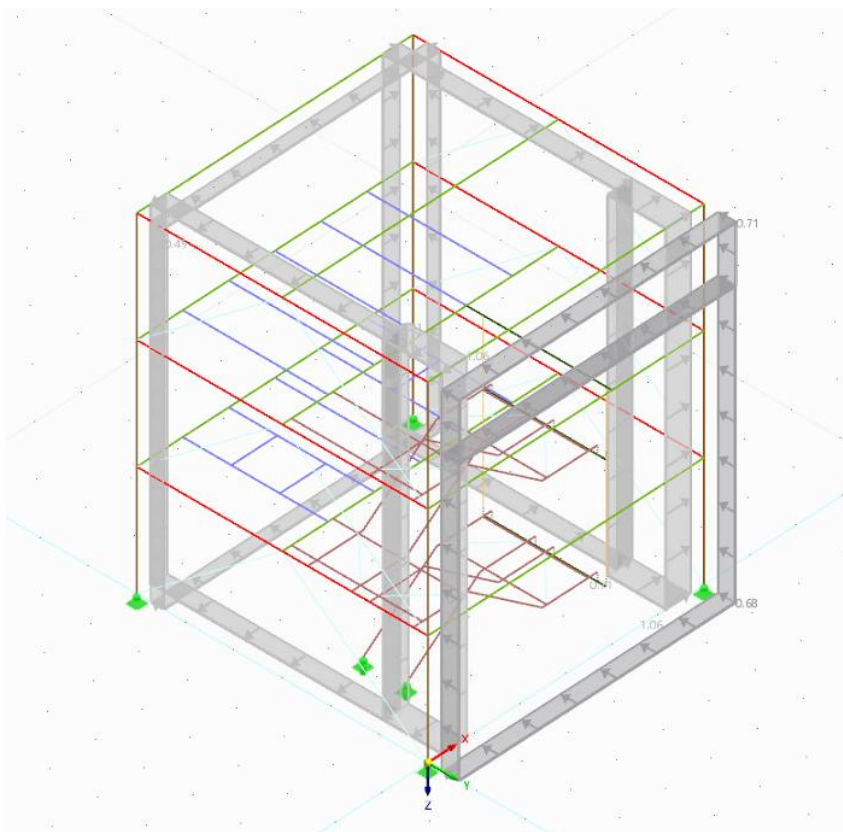
*Karamba3D – Visualisation of wind zones acting in direction +X*

Parallel ▾

*Karamba3D – Visualisation of wind zones acting in direction +Y*



RFEM – Visualisation of wind zones acting in direction +X

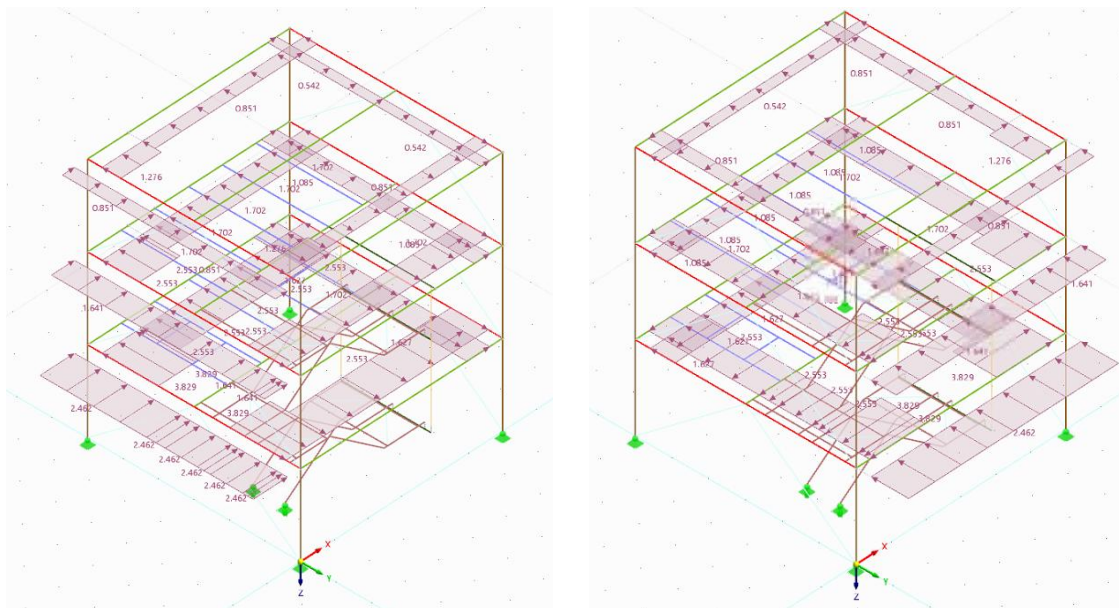


RFEM – Visualisation of wind zones acting in direction +Y (Rhino axis)

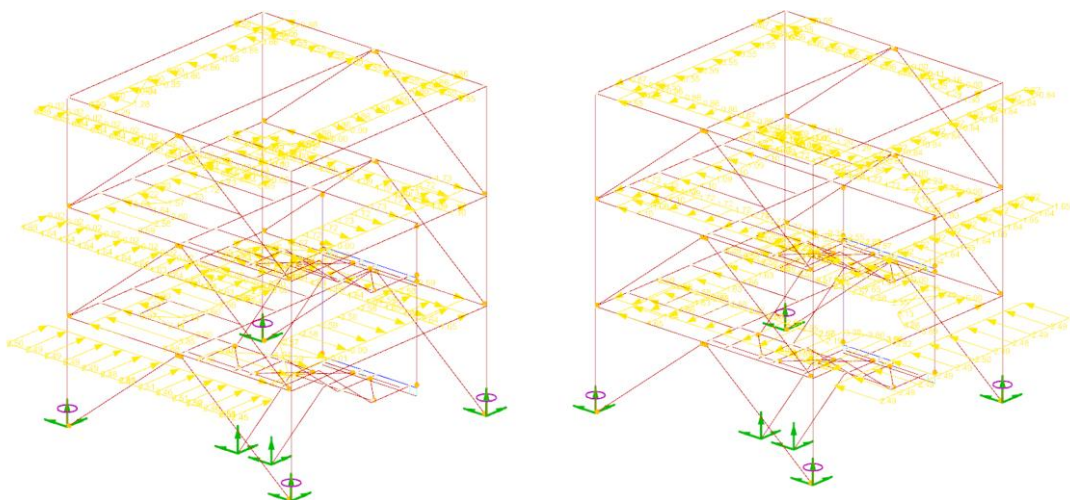
The following images show the visualisation of the member loads determined by RFEM and Karamba3D.

RFEM performs a tributary area algorithm to estimate applied member loads.

Karamba3D performs a direct assignment algorithm closer to the idea of mesh loading the structure directly.



RFEM – Visualisation of applied member loads; Left: direction +X. Right: direction +Y



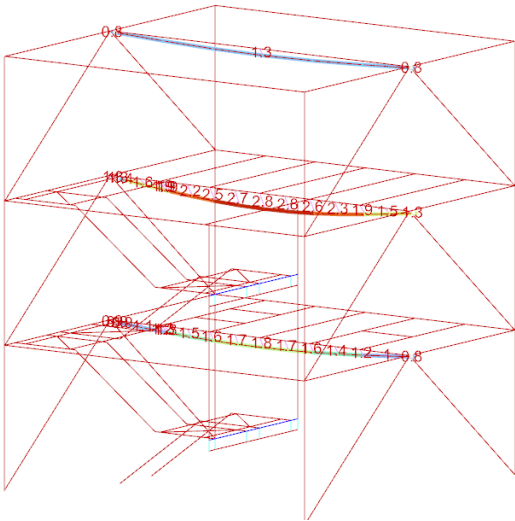
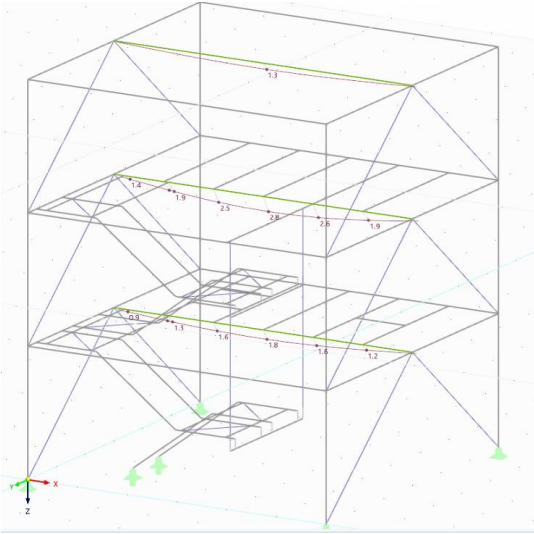
Karamba3D – Visualisation of applied member loads; Left: direction +X. Right: direction +Y

Member Loads						
Direction	Zone	Position	RFEM [kN/m]	Karamba3D [kN/m]	Deviation	Deviation [%]
+X	D	Level 1 Beam	2.462	2.51	0.048	1.9%
+X	D	Level 2 Beam	1.641	1.66	0.019	1.2%
+X	D	Level 3 Beam	0.851	0.86	0.009	1.1%
+X	A	Level 1 Beam	-3.829	-3.87	0.041	1.1%
+X	B	Level 1 Beam	-2.553	-2.58	0.027	1.1%
+X	E	Level 1 Beam	-1.627	-1.65	0.023	1.4%

4 Results Comparison

4.1 Displacements / Deformations

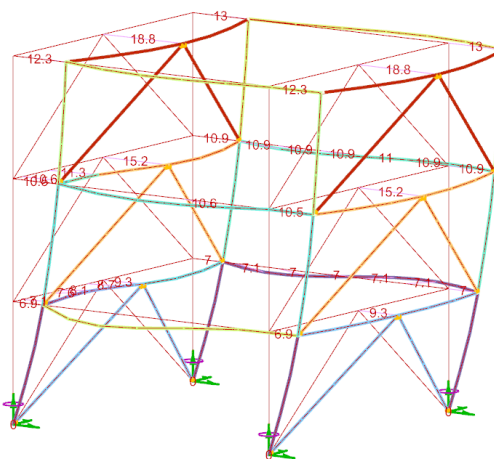
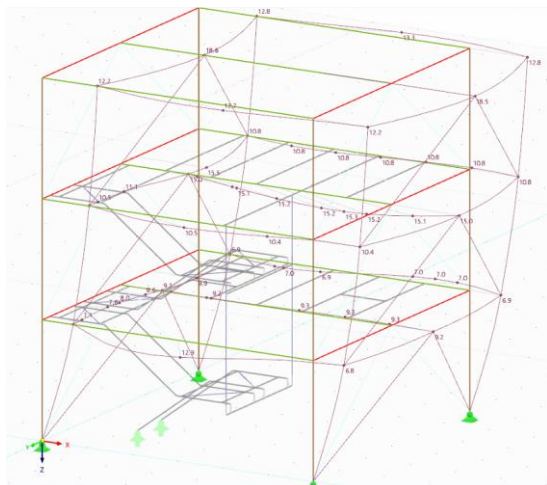
Displacements Gravity					
Load Case	Position	RFEM [mm]	Karamba3D [mm]	Deviation	Deviation [%]
Gravity (1.15 sw factor)	Middle Beam Level 1	1.8	1.8	0.0	0.0%
Gravity (1.15 sw factor)	Middle Beam Level 2	2.8	2.8	0.0	0.0%
Gravity (1.15 sw factor)	Middle Beam Level 3	1.3	1.3	0.0	0.0%



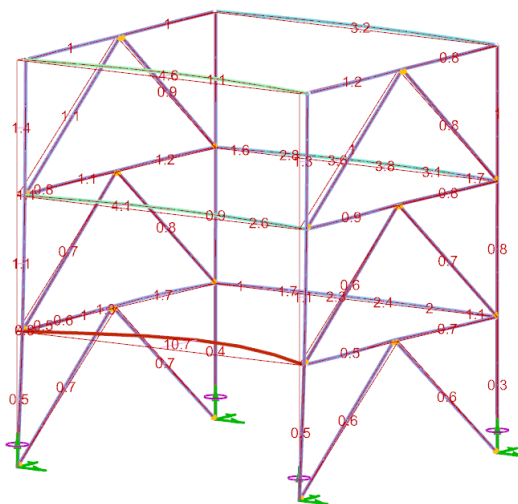
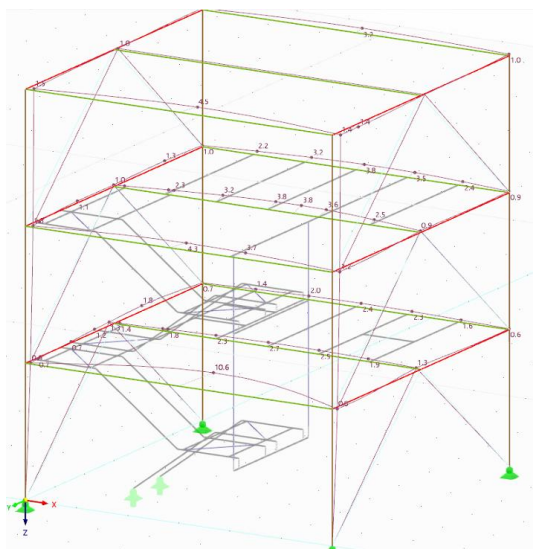
Middle Beams Gravity Deflection – Left: RFEM. Right: Karamba3D (scale = 100)

Displacements Wind

Load Case	Position	RFEM [mm]	Karamba3D [mm]	Deviation	Deviation [%]
Direction +X	Max at top: Middle Beam Level 3	18.6	18.8	0.2	1.1%
Direction +X	Max at top: Stair-side columns	12.2	12.3	0.1	0.8%
Direction +X	Floor-side frame at Level 1	7.0	7.1	0.1	1.4%
Direction +Y	Stair-side beam Level 1	10.6	10.7	0.1	0.9%
Direction +Y	Max at top: Stair-side beam Level 3	4.5	4.6	0.1	2.2%



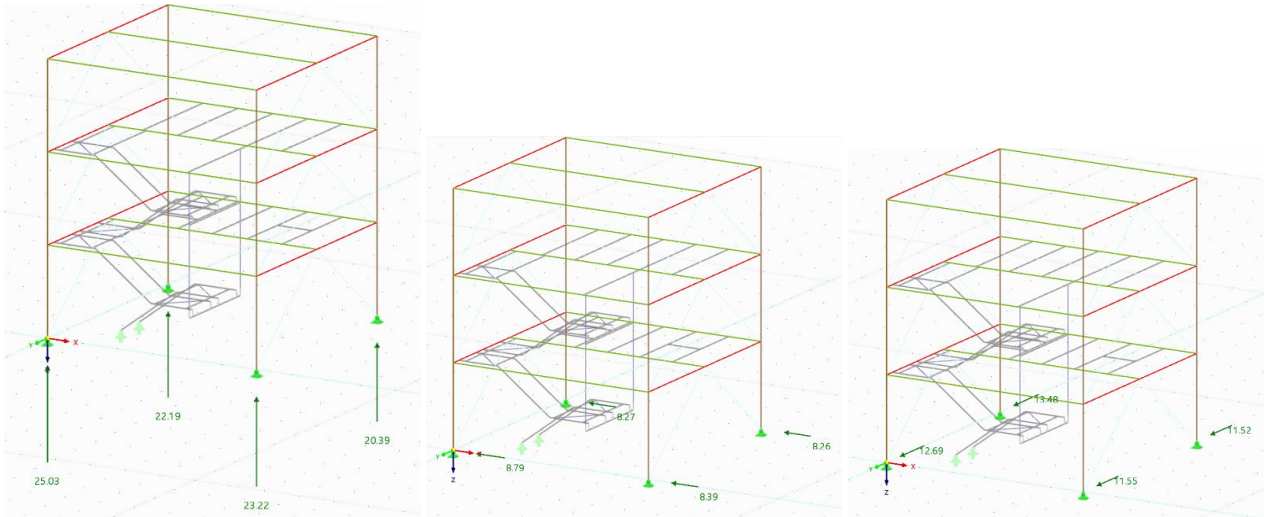
Overall Lateral Deflection Direction +X – Left: RFEM. Right: Karamba3D (scale = 100)



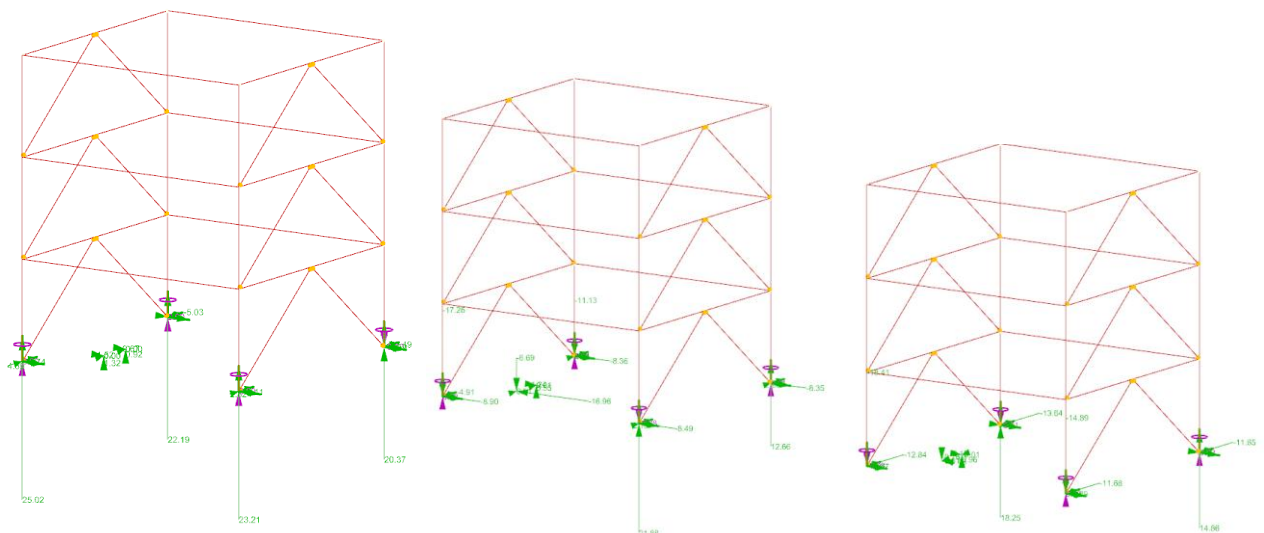
Overall Lateral Deflection Direction +Y – Left: RFEM. Right: Karamba3D (scale = 100)

4.2 Reaction Forces

Support Reactions					
Load Case	Position	RFEM [kN]	Karamba3D [kN]	Deviation	Deviation [%]
Gravity (1.15 sw factor)	Max Rz	25.03	25.02	0.01	0.0%
Direction +X	Max Rx	8.79	8.90	0.11	1.3%
Direction +Y	Max Ry	13.48	13.64	0.16	1.2%



Reaction Forces RFEM – Left: Gravity; Middle: Direction +X; Right: Direction +Y;



Reaction Forces Karamba3D – Left: Gravity; Middle: Direction +X; Right: Direction +Y;

5 Deviation Summary

Maximum deviations across all categories are tabulated below.

Result Category	Max Deviation (%)
Peak Velocity Pressure	Within rounding error
Pressure Coefficients	0.0%
Design Wind Pressure	1.8%
Derived Member Loads	1.9%
Displacements Gravity	0.0%
Displacements Wind	2.2%
Reactions Gravity	0.0%
Reactions Wind	1.3%

6 Conclusion

Overall Validation Result	PASS
Comments / Remarks	Example file ENG002 – Steel Tower contains a C# script which derives Eurocode 1 based wind loads on a rectangular building. The procedure and results of those wind calculations are validated against RFEM Wind Load Wizard for the same conditions. The application of the derived area loads using Karamba3D mesh loads is validated and produces the same kind of member loading on the structure as the RFEM Wind Load Wizard. Finally, the obtained results (reactions, deformations) are closely matched between Karamba3D and RFEM (<2.5% deviation in any scenario investigated).
Author	Dr Dragos Naicu Date: 22.04.2026